



Om Rajendra Kulkarni

B.Tech Electrical Engineer

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Profile

Analytical B.Tech Electrical Engineering graduate with expertise in project planning, technical analysis, and cost optimization through value engineering. Skilled in Artificial Intelligence, Machine Learning, and Robotics, with hands-on experience developing innovative engineering solutions. Adept at cross-functional team collaboration, quality assurance, and delivering results in dynamic engineering environments. Recognized for adaptability, attention to detail, and a commitment to continuous learning.

Professional Experience

Sina Technology - Sina E Bikes Pvt. Ltd.

Electrical Engineer Intern

02/2026 – 05/2026 | Kelwad, India

- Supported battery, motor, and controller integration for e-bike's.
- Conducted electrical testing, documentation, and performance analysis.
- Collaborated with cross-functional teams to improve product quality and reliability.

Virtual Intern – AI & Software Development

Infosys Springboard

10/2025 – 12/2025 | Remote

- Developed an AI-based Resume Matcher and Skill Recommender Tool during the Infosys Springboard Internship 6.0 program.
- Applied software development and machine learning concepts while successfully completing all assigned internship tasks and project requirements.

Emertxe Information Technologies, Bangalore

Embedded Systems Intern (Online)

06/2026 – 06/2026

- Selected for a 4-week Embedded Systems internship focused on core embedded programming and electronics fundamentals.
- Gaining hands-on knowledge of embedded systems, IoT concepts, and real-world engineering practices through structured training and project-based assignments.

Education

B.Tech Electrical Engineering

Sanjivani College of Engineering

11/2022 – 2026 | Kopargaon, India

CGPA - 7.41

H.S.C

Sanjivani Junior College

2020 – 2022 | Kopargaon, India

Percentage - 77.33%

S.S.C

Sainath Madhyamik Vidyalaya

2020 | Shirdi, India

Percentage - 89.40%

Projects

AI-Driven Torque Compensation for Enhanced Stability in Electric Vehicles on Rough Terrain

Designed a CVT system for EVs with MPU & ESP32, integrated an AI-powered dashboard for real-time torque prediction, and prototyped on a toy EV to enhance stability; filed a design patent

Raghava – AI & ML-Based Drone for Surveillance and Mapping

Engineered an AI/ML-enabled drone for surveillance and mapping with IoT integration, incorporating pose-based threat detection and a Smart Rotary Plate for enhanced aerial flexibility.

Skills

Soft Skills

- Adaptability to Dynamic Environments
- Attention to Detail
- Leadership & Team Coordination
- Continuous Learning & Self Motivation

Technical Skills

- Programming & ML/AI: Python, C/C++, Java, C#, MATLAB, TensorFlow, Data Science, Model Deployment
- Data Analysis & Visualization: Pandas, NumPy, Seaborn, Perspective
- Robotics & Embedded Systems: Arduino, Raspberry Pi, IoT, Path Planning, PLC, LiDAR, Sensors
- Simulation & CAD: MATLAB/Simulink, ANSYS, ETAP, CATIA V5, SolidWorks, AutoCAD
- Tools & Platforms: Salesforce, DotNet, RPA, Selenium, Cloud Computing
- Domain Expertise: Industrial Automation, EV Technology, Rapid Prototyping, Value Engineering & Cost Optimization

Professional Skills

- Project Planning & Management
- Cross Functional Team Collaboration
- Quality Assurance & Testing Technical
- Documentation Critical Thinking & Problem Solving

Courses & Certifications

Career Essentials in Generative AI

Microsoft & LinkedIn

Career Essentials in Data Analysis

Microsoft & LinkedIn

Data Scientist Career Track

365 Data Science

Microsoft Azure AI Essentials Professional Certificate

Microsoft & LinkedIn

Simulink Onramp

MathWorks

Learning PCB Design with EAGLE

Autodesk

Power Transmission and Protection Studies – Course Completion Certificate

L&T EduTech

Atlassian Agile Project Management Professional Certificate

Atlassian

VLSI and Embedded System Design Engineer– Course Completion Certificate

L&T EduTech

- Cyber Physical System Engineering
- VLSI Design Engineering

Awards

Certificate of Recognition – TECHgium® 9th Edition

L&T Technology Services Limited

08/06/2026

AI-Enabled Automated Validation of Flow Instrument Installation in 3D Piping Models.

AI-Powered Personalized Grooming Solution – BeYou

Designed an AI-driven facial recognition system with ML-based recommendation engine for personalized grooming and fashion, demonstrating consumer-focused AI applications.

AI-Based Thumper Test for Automotive Components

Developed an AI-powered system using vibration & acoustic analysis with ML models for fault detection, enabling predictive maintenance and enhancing automotive quality assurance

Edubridge Rural - Titan

Contributed to Titan–Edubridge Rural Project by supporting digital skill development and employability training for rural youth.

Patents

AI DRIVEN TORQUE COMPENSATION FOR ENHANCED STABILITY IN ELECTRIC VEHICLES (457698-001)

360 DEGREE CAMERA FOR AI ML BASED DRONE FOR SECURITY PURPOSE (449730-001)

Publications

Artificial Intelligence Applications for Sustainable IoT Systems: A Systematic Review and Future Directions in Environmental Sustainability

International Journal of Environmental Chemistry
Research Paper

Collaborative Human-Robot Teams Enhancing Productivity and Safety in Next-Gen

Research Paper

Torque Management System for Electric Vehicles

Design and Innovation Paper

Languages

Marathi • Hindi • English • German • Japnese

Interests

Tech Blogging • Reading

Top Finalist

Techgium® 8th Edition – National Innovation Challenge by L&T Technology Services

01/12/2024

Recognized for pioneering innovation in industry-driven problem solving.

Top Finalist

DIPEX 2025 (COEP, Pune)

01/12/2024

State-level recognition for autonomous robotics innovation.

Winner

MET Bhujbal, Nashik
AI & ML Based Drone Surveillance System.

2nd Place

Sanjivani National Techfest
AI & ML Based Drone Surveillance System.

2nd Place

Matoshri Institute Techfest
AI & ML Based Drone Surveillance System.

Certificate of Recognition – TECHgium® 9th Edition

L&T Technology Services Limited
05/06/2026

AI-Driven Automated Testing Framework for OTT and Digital TV Ecosystems.

Organisations

Sanjivani College of Engineering

Internship & Placement Coordinator

Team Raghava Innovations

Founder & Lead Innovator
Driving AI/ML and EV-based projects for national-level competitions

Sanjivani College of Engineering

Hackathon Coordinator